Session Advanced Functions – Create IPO Chart and code for each problem below.

1. The input consists of quantity, price, and discount rate. Use a function to compute the <u>discount amount and the discounted price</u>. Then display these values in the main along with the quantity and price. (The function should return both the discount amount and the discounted price.

INPUT	PROCESS	OUTPUT

QTY
PRI
DIR
	Get qty, pri, dir
	
	Call compute_disc
	    Input: pri, qty, dir
	    Return/ update:
	    discamount , discprice
	
	Extprice = qty * price
	discamount = extprice * dir
	discprice = extprice - discamount
		QTY
		PRICE
		DISCAMOUNT
		DISCPRICE

2. Enter the student's last name and 3 exam scores. Use a function to compute the average and total points. This functions should return both total points and exam score. Display student last name, total points and average exam score.

INPUT	PROCESS	OUTPUT

Lname
S1
S2
S3
	Get lname, s1, s2, s3
	
	Call average_points
	    Input: s1, s2, s3
	    Return: total, avgscore
	
	total= s1+s2+s3
	avgscore= total/3
		Lname
		Total
		avgscore

3. Produce a sales report. Input salesperson last name and sales. Write a function that compute commission which is 10% for sales over $100, 000 and 5% for sales at or under $100,000. The function should also computer next year's target which is 5% of the sales. This function should return both commission and next year's target. Display salesperson name, commission and next year's target.

INPUT	PROCESS	OUTPUT

Lname
Sales
	Get lname, sales
	
	Call nxtyeartarget
	        Input: sales
	        Return:
	        Commission, nxtyeartarget
	
	if sales<= 100000:
	        commission = .05
	    else:
	        commision = .10
	
	commisionreturn = sales * commision
	nxtyeartarget = sales * 0.05
		Lname
		commisionreturn
		nxryeartarget

4. Enter bowler last name, 3 game scores and handicap. Write a function to compute average score and average score with handicap. Back in main, display last name, average score and average score with handicap.

INPUT	PROCESS	OUTPUT

lname
gs1
gs2
gs3
handicap
	Get lname, gs1,gs2,gs3, handicap
	
	Call game_total
	        Input: gs1, gs2, gs3
	        Return: total, avgscore
	Call handicap_total
	        Input: gs1, gs2, gs3, handicap
	        Return: handicapscore
	
	total= s1+s2+s3
	avgscore= total/3
	avgwhandicap = handicap_total/3
		Lname
		avgscore
		avgwhandicap

5. Allow the user to enter quantity of an item and unit price. Write a function to compute total (qty * unit price) and tax (7% of total). Demonstrate your knowledge of global variables by making total and tax global in scope. Display total and tax in main.

INPUT	PROCESS	OUTPUT

QTY
UPRI
	# Global variables
	total = 0
	
	def compute_totals(qty, price):
	        Input: global total, tax
	        Return: total, tax
	
	def():
	      qty (float(input(“Input the quantity of the item:”))
	      upri (float(input(“Input the price of the item:”))
	       compute_totals (qty, price)
	
	total = qty * price
	tax = total * 0.07
	total_cost= total + tax
	
	Print (“Price of items:”, total)
	Print (“Total tax of the items:”,total_tax)
	Print (“Overall cost:”, total_cost)
		Total
		Tax
		Overall cost